EdgeAD: Unsupervised Learning Model Based on Prior Knowledge Enhanced Image Anomaly Detection of Heavy Railway Freight Cars

Weiyu Zhang, Hongmei Shi, Ji Qiu, Zujun Yu, *Member, IEEE*, and Jianbo Li

***Abstract*— Trouble of moving freight car detection system (TFDS) is an important safeguard measure for the operation of railway freight cars. For few- and zero-shot faults that seriously affect traffic safety, it is a tough problem to realize automatic TFDS image recognition. In recent years, anomaly detection (AD) models based on unsupervised learning have provided a solution to this problem, but the current accuracy (ACC) and detection efficiency still need to be improved. The current AD model mainly reconstructs the image with the global information of the whole image and lacks the effective employment of focused information. This article proposes an AD algorithm with prior knowledge to enhance the ACC and reduce the time cost. First, an edge information-guided autoencoder image reconstruction algorithm is designed using edge information of TFDS images as the prior knowledge, which achieves improved performance by eliminating invalid information. Second, an edge-weighted loss is proposed to enhance the constraining effect of edge information during the training process. Finally, an empirical mask mechanism is proposed, which is guided by the prior knowledge of the anomaly characteristics. Validated on the real-world TFDS-based dataset, the proposed model improves the detection performance by 21.8% compared to the variational autoencoder (VAE) model. The proposed model is also extensively and comprehensively quantitatively evaluated against mainstream industrial image AD models, and the proposed model has better ACC and detection speed in TFDS image AD.**

***Index Terms*— Loss function, prior knowledge, unsupervised anomaly detection (AD), variational autoencoder (VAE).**

I. Introduction

IN RECENT years, with the development of image equipment and communication technology, the research of combining image technology with railway fault detection has been deepening. Trouble of moving freight car detection system (TFDS) uses multiple groups of high-speed industrial cameras installed on the rail side and the rail between to take dynamically captured images of railway freight cars in operation. The collected images are transmitted to the train inspection center through the network, and manual fault recognition of freight cars is carried out by inspectors.

The current heavy railways in China are characterized by large capacity, long formation, dense passing, short station stopping time, and so on. This manually based TFDS dynamic inspection method faces the challenges of short operating intervals, high work intensity, and high influence of human factors on the inspection quality. The study of intelligent recognition of railway images is considered a way to solve these problems.

Previously, in the research of image intelligence recognition, the solution idea of supervised fault detection is usually used to realize the recognition of certain faults. However, in the problem of fault detection in freight car images, there are challenges such as low proportion of fault images, asymmetry of data samples, and difficulty in acquiring fault samples. Besides, faults that seriously endanger the safety of railway freight cars rarely occur, so it is difficult to collect the samples.

The performance of traditional fault detection models depends on the quantity and quality of fault samples. Without a large number of representative fault samples to train the model, the model cannot learn the characteristic information of the faults adequately, resulting in the inability of the model to achieve high-precision fault detection in practical applications.

In recent years, anomaly detection (AD) has achieved several significant advances and results in engineering applications. The AD objective is to find outliers, abnormal patterns, or abnormal behaviors in a dataset. Compared to supervised fault detection, which has a specific fault as its detection target, unsupervised AD based on the target can be more flexible and adaptable to a variety of complex engineering application contexts.

The research on intelligent recognition algorithms for railway vehicle images can be mainly divided into two stages.

The first stage is based on image processing and traditional machine learning recognition methods. Liu et al. [1] realized bearing weight saddle positioning based on the Hough transform and detected bearing weight saddle misalignment fault based on the symmetry method. Cao et al. [2] designed a weighted edge sparse embedding classifier for railway freight car brake cylinder detection. Sun et al. [3] proposed a fault detection model based on image segmentation and shape

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matching to realize automatic detection of air braking system, bogie block keys, and fastening bolts. Lu et al. [4] used time-scale normalization to achieve accurate registration of target images and proposed an image difference method to locate the position of potential fault regions. Karakose et al. [5] combined canny edge detection, Hough transform, and decision tree classification to detect track components and monitor track line conditions. The above research solves some problems of image detection, but the recognition accuracy (ACC) of methods based on image processing and traditional machine learning will decrease significantly due to complex environments and diverse fault types.

With the development of deep learning technology, more and more researchers are devoted to using advanced deep learning models to improve the efficiency and ACC of fault detection, which is the main research aspect of the second stage.

Convolutional neural network (CNN) is a widely used deep learning model. Sun et al. [6] used the coarse-to-fine CNN fault identification method to simultaneously detect the target area of the side frame key (SFK) and shaft bolt (SB). Another CNN model for multiple fault determination is built to identify four typical faults in the target area. Fu et al. [7] used CNN to realize bearing positioning and proposed an improved U-Net method for bearing oil leakage defect recognition.

Researchers have also explored many improved CNN models for the detection needs of railway images. Xiao et al. [8] proposed an improved Mask-RCNN model to identify missing faults of pins, bolts, and rivets. On this basis, Chang et al. [9] developed a fault diagnosis method combining OSCN and RIN for the state diagnosis of key components of high-speed running trains. Zhang et al. [10] designed FDNet to detect the fault detection of three typical train components, namely, the cutoff stopper door handle, the upper lever, and the locking plate, to achieve efficient train image fault detection. To improve the detection effect of small-sized component faults, Ye et al. [11] embed feature fusion, multibranch dilated convolution, and the squeeze-and-excitation block into the model. The abovementioned methods provide a basic framework to accomplish object detection. However, CNN networks only consider feature information from local regions in the convolution operation, which may result in the features extracted by the network being too localized to capture the global feature relationships. Moreover, as the depth and width of the network increase, the number of parameters in the network also increases, which can lead to an overly complex network, increasing the time for training and the cost of computation.

To solve these problems, some researchers have begun to try other architectures based on supervised learning to improve the detection performance. Zhang et al. [12] proposed a heterogeneous self-distillation framework to realize the fault detection of four key components: bogie block key, cutoff cock, dust collector, and brake beam fastening bolt. The knowledge distillation strategy and lightweight backbone network were used to improve detection ACC and efficiency. Hütten et al. [13] applied a VIT-based classification model to damage detection of railway freight car floors and compared it with a traditional CNN method. The experimental results validate the superiority of ViT in damage detection tasks. On the other hand, Zhang et al. [14] designed a lightweight anchor-free framework MS FTI FDet to reduce the number of parameters of the model and realize the fault detection of railway freight car components of different sizes.

In summary, the current research mainly relies on supervised learning. This approach requires a certain number of labeled fault samples so that the model can learn and identify fault features. In practice, the low fault frequency of railway freight cars makes it difficult to obtain enough fault samples, which is time-consuming and laborious to collect a large amount of annotated data. This problem limits the performance of supervised fault detection methods for railway freight cars.

Unsupervised AD provides a solution to this problem. AD algorithms based on unsupervised learning include probabilistic model methods, distance-based models, dictionary learning methods, change detection frameworks, and reconstruction-based methods [15].

Applying unsupervised learning to the field of industrial images is one of the hot research directions in recent years. Unsupervised image AD algorithms automatically identify defects and anomalies through image reconstruction. The anomaly method based on learning feature representations of normality is one of the significant methodologies for AD [16], mainly represented by generative adversarial networks (GANs) [17] and variational autoencoders (VAEs) [18].

However, there is still little research on applying anomaly image detection methods to TFDS images. Currently, there have been advances in unsupervised learning research for train image detection. Neto et al. [19] applied a combination of VGG19 deep CNN and semisupervised DRB classifier to evaluate the condition of freight car bogie springs. Kang et al. [20] developed a deep architecture for surface defect detection of high-speed railway insulators using denoising autoencoders. Zhang et al. [21] proposed a detection algorithm based on GAN of image similarity for defect detection of multiple types of high-speed train components. Their research demonstrates the potential of unsupervised learning in train images.

Through the research on related work, it reveals that the following conditions hold.

1) Currently, most of the research on automatic identification of TFDS is based on supervised learning. However, TFDS has the characteristics of low fault frequency and few fault samples, and the supervised learning model cannot ensure high ACC. In addition, existing research and detection methods usually only focus on a single or a few kinds of faults, which cannot deal with a wide variety of TFDS faults.
2) There are few studies on unsupervised AD for TFDS. AD algorithms in the field of industrial images usually rely on the global information of the image. However, TFDS images are grayscale images, and conventional algorithms are ineffective. In addition, many TFDS faults can be guided using edge information as prior knowledge.

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3) In addition, practical applications in this field require improving the efficiency of the model. Compared with models that utilize global information to reconstruct images, guiding the model with edge information as a priori knowledge reduces the involvement of redundant information and achieves a significant improvement in the efficiency of the model.

This article proposes an unsupervised learning-based AD model to solve the problem of intelligent identification of TFDS faults. Considering the lack of color information in the image of TFDS and the fact that TFDS anomalies can cause deformations that change the edge information characteristics of the image, it proposes an AD model based on the guidance of prior knowledge enhancement. The main contributions of this article are the following four aspects.

1) A novel unsupervised AD model is presented in the context of TFDS AD applications, in which an image generation algorithm guided by edge information as priori knowledge is proposed.
2) In the construction of the edge-weighted loss function, the edge-guided loss, structural similarity (SSIM), and gradient magnitude similarity (GMS) are considered to enhance the model capability to leverage edge information and reinforce the role of edge information as a constraint in the training process of the model.
3) An adaptive optimization process of the mask is designed, in which the size of the mask can adjust according to the size of the target component, and its effectiveness in applying it to TFDS images is experimentally verified.
4) The TFDS anomaly image dataset is established, which is utilized to verify and enhance the scalability and universality of the detection algorithm through process optimization.

The remaining sections of this article are as follows. Section II illustrates the framework of the EdgeAD model. Section III provides the experimental setup and the main results. Section IV discusses the performance of the proposed model and the impact of the key parameters in the model and compares it with the state-of-the-art methods. Finally, in Section V, the conclusion is drawn and summarized.

II. Methods

The process of TFDS is shown in Fig. 1. The train inspector uses the software platform through the TFDS system and manually browses the picture of the way to fault analysis and judgment.

The application context of this article is the AD of TFDS images. Upon analyzing the TFDS images, it was discovered that anomalies occur infrequently, making it difficult to acquire anomaly data. Moreover, anomalies may appear at any location on railway freight cars, and their type may vary. When using supervised learning methods for AD, the challenge lies in constructing a dataset of well-labeled anomaly samples, which requires a large amount of such data and presents difficulties in acquisition.

In contrast, unsupervised learning methods model data based on its inherent characteristics, eliminating the need for prelabeling anomaly data. In situations where acquiring anomaly data is problematic, the model can adaptively identify anomaly based on data distribution characteristics. Since unsupervised learning methods do not rely on a priori labeling information, they can flexibly process data type and distribution. This enables them to adaptively capture data distribution features and effectively detect anomalies of various types, demonstrating better robustness. Therefore, employing unsupervised AD can more effectively detect anomalies in TFDS images.

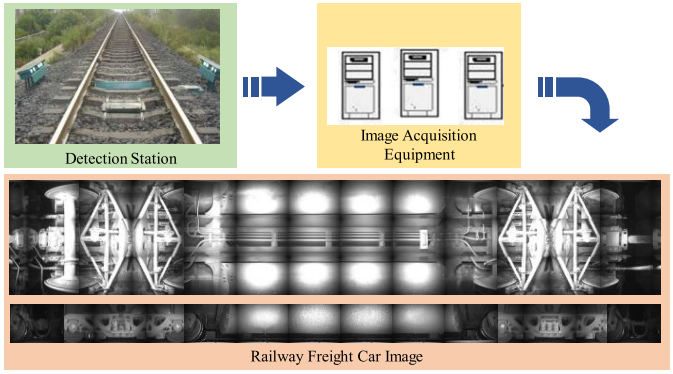


Fig. 1. Trouble of moving freight car detection system.

A. Dataset

We collected TFDS images from a TFDS detection station on a freight railway line in operation and obtained original and fault images of freight cars. Establish a TFDS image database, collecting over 800 000 original and 3000 fault images covering over 100 faults.

In the background of freight car images, anomaly usually refers to the appearance of unconventional situations or elements in the image. These anomalies indicate that the railway freight cars in the image have malfunctioned or have safety issues.

Fig. 2(a)–(d) shows some examples of anomaly images, showing four types of faults: component-missing-type fault, breaking-type fault, deformation-type fault, and foreign object. It can be seen that when an anomaly occurs, the edges of the image change visibly compared to the normal condition.

This article collects the TFDS inspection data of an inspection center from February 2022 to February 2023. Through the statistics of TFDS fault records, the faults are divided into eight types according to deformation type faults, bearing lubrication splashings, foreign objects, damage-type faults, component-missing-type faults, scraping-type faults, overlimit type faults, and breaking type faults, and the ratio of occurrences of each fault is plotted in Fig. 3.

Therefore, from the fault proportion shown in Fig. 2, TFDS images have their own unique characteristics. Upon analyzing TFDS images, it was observed that TFDS images lack color information, and most faults have no significant effect on texture information after the occurrence of faults, so these two types of feature information are of little value to be utilized in TFDS fault detection.

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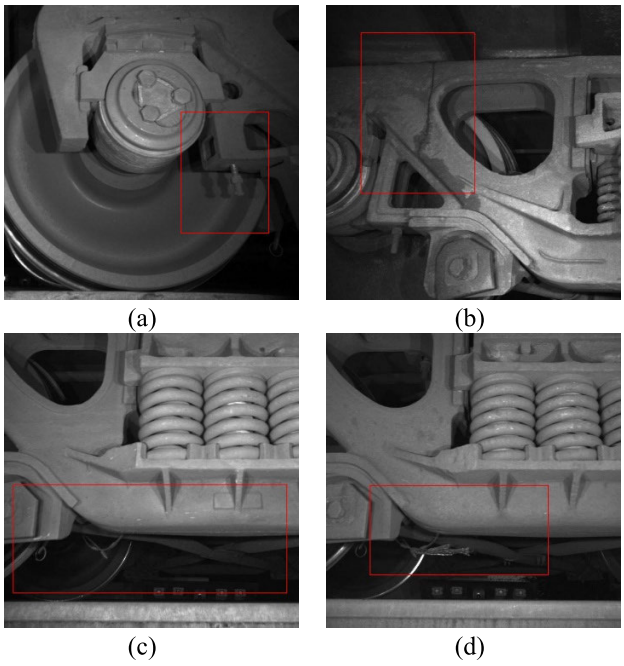


Fig. 2. Anomaly example images. (a) Component-missing-type fault. (b) Breaking type fault. (c) Deformation type fault. (d) Foreign object.

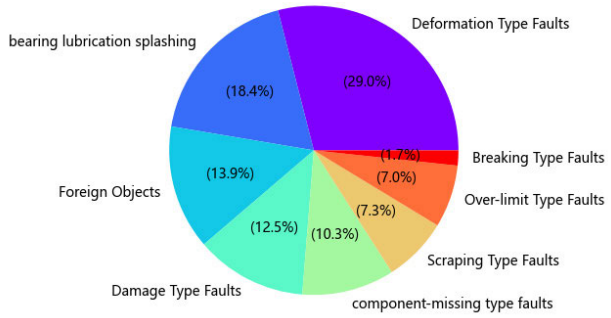


Fig. 3. Fault percentage pie chart.

On the other hand, all anomalies will affect the image edge information. Focusing on the change of edge information can meet the needs for intelligent detection of TFDS. In addition, using only the edge pixels as a benchmark for the measurement of the reconstruction constraints will save a lot of time compared to targeting the global pixels, which has a positive impact on improving the running speed of the model.

B. Edge Information-Guided Image AD

Currently, the autoencoder model usually consists of two components: an encoder, which learns input data features and compresses them into a low-dimensional representation, and a decoder, which reconstructs the original data form from these representations. During the detection phase, the VAE-based AD model uses the calculation of the difference between the reconstructed image with the original image to achieve the detection of anomaly.

However, this method requires comparing all pixels of the reconstructed image with the original image, which makes improving real-time performance challenging. At the same time, it also needs to consume a lot of computing resources to achieve higher detection ACC.

Therefore, to improve the detection efficiency and ACC of the model, it is necessary to enhance its ability to process the information of the dataset. Consequently, further in-depth research is required to improve and optimize the TFDS image feature extraction part within the AD model framework.

Models with enhanced detection performance using prior knowledge have been applied [22]. Based on the analysis of TFDS images, the importance of edge information can be seen.

Our proposed EdgeAD presents the idea of using edge information to guide the reconstruction. The focus of EdgeAD work is to detect anomalies in the image. In the autoencoder-based AD model, the role of the VAE is to generate the reconstructed image from the input raw image. In the training phase, the training set has a large number of raw images, of which normal images account for a large proportion. During the training phase, the model gradually learns how to generate reconstructed images that are closer to normal. In the detection phase, the model will reconstruct the input TFDS image into a near-normal image. For normal images, there is not much difference in the reconstruction results. However, when the model encounters abnormal images, there is a noticeable difference between the input and output of the model. Through this method, AD can be achieved.

The main architecture of the proposed model is shown in Fig. 4. The main architecture of EdgeAD algorithm consists of two parts: image reconstruction module and edge information extraction module. The backbone part of the model is a network structure based on U-net [23], as shown in Fig. 5.

C. Edge Information Extraction Module

AD algorithms applied in the industrial image field usually use the overall hidden patterns of the image, texture features, color information, and so on, as feature information to guide the model reconstruction and identification of anomalies. Since TFDS images are grayscale, texture and color are not obvious and plentiful.

In our previous analyses, we found that the occurrence of faults in TFDS images will reflected in the edge information of the images. Focusing on the changes in the edge information of the image can help the model to better identify anomalies in the physical boundaries and component contours in the images.

Therefore, the edge information extraction module needs to be added to the model to help the model make effective use of the edge information. The edge information extraction module can work as a feature extractor to assist the model in better capturing the edge information in anomaly images. The working principle of edge extraction is based on the grayscale variation of an image, and the edge location is determined by detecting regions with drastic changes in pixel values by performing difference operations or filtering on the pixel points of the image.

Among the commonly used edge detection algorithms, the Canny algorithm is a widely used edge detection algorithm for industrial images. It has good edge localization ACC and anti-noise ability to meet the needs of edge information

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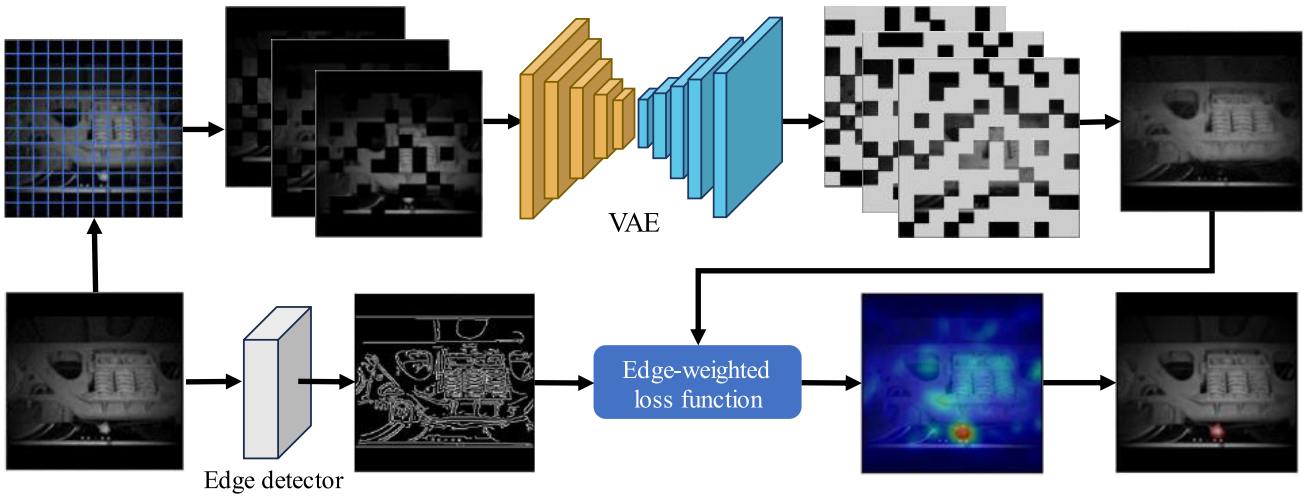


Fig. 4. Model architecture diagram.

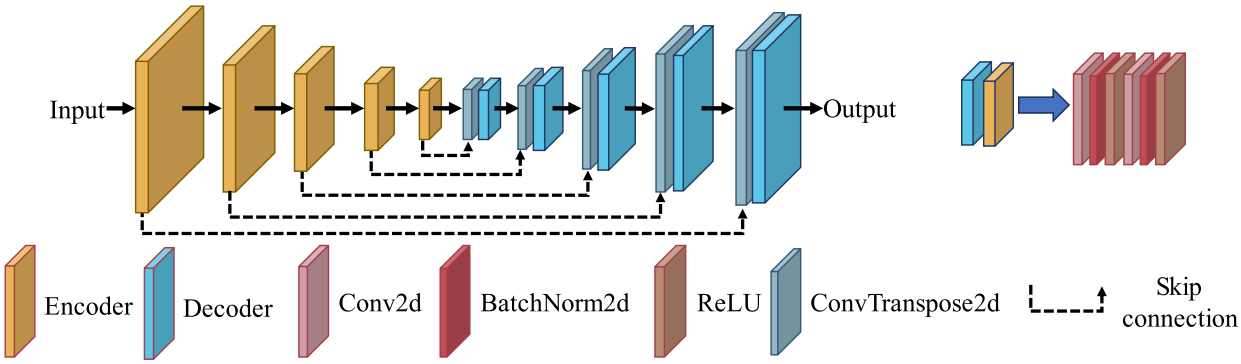


Fig. 5. U-net backbone.

extraction in TFDS images. Therefore, the edge information extraction module in this article is designed based on the Canny algorithm.

When comparing the difference between the reconstructed image and the original image, the edge information is involved in the loss calculation of the loss function, which can constrain the reconstruction of the model to be optimized in the direction of more ACC and clarity, thus improving the ACC and robustness of AD. At the same time, by adding edge information to the image reconstruction process, the model can detect the edge changes more sensitively in the anomaly image and improve the sensitivity of AD.

The role of the edge information extraction module is to extract the edge information of the image by the edge detection algorithm before the original image enters the image reconstruction module. The obtained edge information is involved in the model training with the weight of the loss function. It is used to guide the model to better recognize the changes caused by anomalies on the edges of the components in the image and complete the detection of anomalies.

Compared to models that use global information for image reconstruction, edge information can enable the model to more accurately focus on changes in key image features, effectively saving the parameters of the model calculation.

D. Edge-Weighted Loss Function

In order to improve the utilization of edge information. It is necessary to improve the loss function of the model. By designing a new loss function, the model can be constrained to utilize the edge information more effectively.

The loss function plays an important role in the training effect of the image generation model and the subsequent AD performance, the loss function is mainly used to measure the gap between the generated image and the real image. In the training process, the loss function can constrain the model to extract image features so that the model generation is more in line with the expected results and constantly help the model to optimize the parameters so that the model can generate a generated image similar to the real image. On the other hand, the loss function can also be useful in the AD stage.

Typically, autoencoder-based anomaly models often use $L2$ loss, also known as mean square error (mse). $L2$ loss mainly takes into account the information of the full image. In the background of this article, the model needs to extract and utilize image edge information. Therefore a new edge extraction function needs to be designed to constrain the model to pay more attention to edge information. In addition to this, outliers are generated when extracting edge information. $L2$ loss is very sensitive to these outliers, and using $L2$ loss to constrain the model generates overfitting [26].

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The Smooth $L1$ loss used in the fast-rcnn framework [27] not only combines the characteristics of $L1$ loss (absolute difference) and $L2$ loss (squared difference) but also reduces the influence of outliers on the loss function

$$L_{\text{Smooth}L1}(I_X, I_Y)_{i,j} = \begin{cases} 0.5\left(X_{i,j} - Y_{i,j}\right)^2, & \left|X_{(i,j)} - Y_{(i,j)}\right| < 1 \\ \left|X_{(i,j)} - Y_{(i,j)}\right| - 0.5, & \left|X_{(i,j)} - Y_{(i,j)}\right| \geq 1 \end{cases} \tag{1}$$

where I_X and I_Y represent the input image and output image, respectively, while $X_{(i,j)}$ and $Y_{(i,j)}$ represent the pixels on the input image and output image.

Therefore, edge-guided loss can be constructed by combining edge information. The expression is as follows:

$$L_E(I_X, I_Y) = \frac{\sum_{i=1}^{W}\sum_{j=1}^{H} E_{i,j} \cdot L_{\text{Smooth}L1}(I_X, I_Y)_{i,j}}{WH}. \tag{2}$$

E is the weight obtained from the edge information of the input image; the weights corresponding to the edge pixels are 1 and the others are 0.

Single loss function can have advantages in processing specific information, but it reduces the robustness of the model. The combined loss function can provide richer gradient information and help the model find the optimal solution faster in the training process. Therefore, when using the edge-guided loss function, introducing other loss functions can effectively improve the performance.

GMS deviation (GMSD) [28] is a full reference evaluation of images method, which evaluates the degree of difference between images by calculating the similarity of the gradient magnitude of the images. GMSD not only provides information about the overall quality of the image but also captures local details and structural variations. GMSD is defined as

$$L_G(I_X, I_Y) = \frac{1}{WH}\sum_{i=1}^{W}\sum_{j=1}^{H} 1 - \text{GMS}(I_X, I_Y)_{i,j}. \tag{3}$$

SSIM loss is based on the SSIM index [29] as a loss function that reflects the SSIM between the generated image and the original image. SSIM is defined as

$$L_S(I_X, I_Y) = \frac{1}{WH}\sum_{i=1}^{W}\sum_{j=1}^{H} 1 - \text{SSIM}(I_X, I_Y)_{i,j}. \tag{4}$$

The edge-weighted loss function is composed by using the edge-guided loss function along with SSIM, which can focus on the structural information, and GMS, which focuses on the gradient information of the image. The expression is as follows:

$$L = L_E + \lambda_G L_G + \lambda_S L_S \tag{5}$$

where λ_G and λ_S are the weighting coefficients of SSIM and GMS, respectively.

The experimental loss function curves are shown in Fig. 6.

As can be seen from the figure, the edge-guided loss converges quickly converges to stability. The strategy of focusing on key information allowed the model to learn the patterns and regularities in the training data relatively quickly.

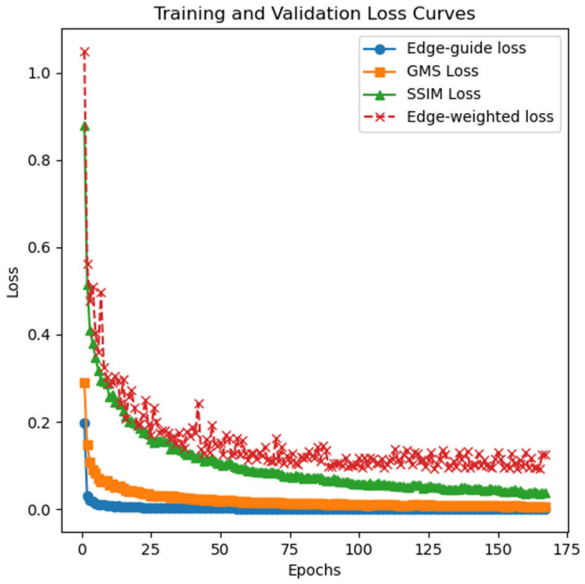


Fig. 6. Loss curve.

GMS also stabilized in the middle. SSIM converged relatively slowly, but there was also a clear convergence trend. The Edge-weighted loss function has small fluctuations during the descent process but eventually stabilizes. From the loss curves in the figure, it is obvious that the model effectively learns the key information of the data.

E. Mask Mechanism

Mask mechanism is often applied in unsupervised AD algorithms, which can effectively improve the ability of the model to generate normal images. Most of the common mask mechanisms are designed to improve the model's ability to restore normal images by selectively ignoring part of the pixels of the original image. Especially in the application background of AD, mask operation on the input image can effectively exclude such interfering factors as anomalies or noise so that the model can learn as much as possible about the image features of the normal image, effectively improving the model's ability to recognize anomalies, and then improve the ACC of the model's detection of anomalies.

However, excessive use of the mask operation may lead to model overfitting [24], which is detrimental to the generalization ability of the model. Also in some cases, the mask may mask potential anomalies, leading to a decrease in the model's ability to detect anomalies.

This article introduces the idea of the mask mechanism, using the reconstruction-by-inpainting (RBI) mask mechanism [25]. The original input image I_X is segmented into a rectangular grid consisting of $k \times k$ pixel-sized blocks, and then, the rectangular grid is divided into n nonintersecting subsets. The pixel blocks belonging to each subset are subjected to the mask operation to obtain a set of n input images, which are sent to the model for reconstruction to generate n reconstructed images, and each reconstructed image rebuilds the region that has been masked in the corresponding input image. This set

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of n reconstructed regions is combined to obtain the complete reconstructed image I_Y.

Different from the previous simple mask operation on the image, the benefit of this mask mechanism is that it can be processed in parallel, and after the complete input image is partitioned into n subsets, a single subset can be processed in parallel, which improves the efficiency of image reconstruction and at the same time reduces the difficulty of reconstruction of large-size images. Most importantly, in the field of AD, it can be done to shield out the anomalous regions reduce the model's opportunity to learn the anomalous regions and improve the model's ability to generate normal images.

The whole mask mechanism, which plays an important role in this part of the model's reconstruction to generate an image, is mainly used to randomly generate blocks of pixels with side lengths of K pixels and randomly mask out some of the pixels in the original image by means of the generated blocks of pixels. The mask mechanism reduces the chances of the image to learn the abnormal regions and thus improves the ability of the model to generate a normal image.

An image with some missing parts is generated as an input for the training of the image generation model, which helps the model to learn to compensate for the missing parts of the image and generate a generated image close to the normal image. The trained model, when encountering anomalous image inputs, generates an image that is also close to the normal image, where the input images I_X and the reconstructed image I_Y differ in the location of the anomaly, thus enabling the detection and localization of the anomaly.

Ideally, a single-pixel block should be able to cover the pixel region where the anomaly occurs in the anomaly image. However, the anomalies appearing in TFDS images are characterized by a large variety, different sizes of anomalous regions, and variable locations of anomaly occurrence, and selecting pixel blocks with appropriate locations and sizes can better help the model to generate high-quality images.

Due to the mask mechanism, we can generate mask pixel blocks at random locations on the input image to cope with the randomness of the anomaly locations. The size of the mask pixel blocks is determined by the K-value, so the K-value needs to be selected to fit the size of the anomaly region of the image as much as possible.

For TFDS images, the anomalies mainly happen in the region where the components are located, so there is a strong correlation between the anomaly region and the component region. Therefore, the pixel size of the widget can be used for selecting the K-value. By analyzing the pixel area of the component, the ratio of the component's pixel size to the whole image can be obtained, as shown in Table I.

Since the images in this article are input into the model for training at 256 × 256 pixel size, when the image edge length is 256 pixels, it can be found that the edge lengths of these regions fall in the interval [8], [32]. Therefore, we hypothesize that using the edge lengths of the pixel blocks, i.e., the K-values, should also be in this interval to produce better training and detection results. In Section IV, we have designed experiments to verify the impact of K-value on detection performance.

TABLE I
PROPORTION OF COMPONENTS IN THE IMAGE

Component	Component length proportion	Component length with image resolution of 256 * 256 pixels
Side frame key	0.056	14.250
Floor	0.071	18.279
Cross-braced rod	0.069	17.689
Locking plate	0.117	29.952
Bolster spring	0.123	31.545
Brake shoe kay	0.038	9.827

III. EXPERIMENTAL VERIFICATION

A. Experimental Settings

Experiments were conducted on the collected TFDS anomaly dataset to verify the detection performance of the model. In order to facilitate the training of the model as well as to refer to the size of the images in the MVTec anomaly detection (AD) dataset, the images collected by the TFDS system are padded so that the image with the size of 1400 × 1024 is padding to a square size of 1400 × 1400. The square size of the input image facilitates the training and evaluation of the model, is more suitable for matrix operations and processing, and is more conducive to the design of our model structure and parameters. At the same time, this processing method can take into account other different sizes of images to ensure the consistency of the dataset, improve the compatibility of the model for the image input size, and ensure that the model has a higher detection performance and generalization ability in practical applications.

The code was run on a computer with 32-GB memory, the GPU was RTX-2060, and the processor was Intel Core i9-10 900. The running experiment is Windows 10, CUDA 11.6. The PyTorch framework is used as the framework. Because of the GPU memory limitation, the model can only be input at a maximum image size of 256 × 256 pixels for training and validation.

B. Evaluation Metrics

In order to evaluate the performance of the proposed EdgeAD model, we compare EdgeAD with GANomaly [30], f-AnoGAN [31], and VAE, which are commonly used methods in the field of industrial image AD. For the selection of evaluation metrics, the method was evaluated on two scales: image-level AD and pixel-level anomaly localization, using pixel- and image-level receiver operating characteristic area under the curve (ROCAUC) as evaluation metrics. ROCAUC is a standard evaluation metric in image and video AD efforts [32], [33], [34]. In addition, the pixel-based ROCAUC score is used as an anomaly localization metric to perform a more detailed evaluation of the anomaly localization performance of the model.

Meanwhile, in practical application scenarios, the running speed of the algorithm is also an important performance

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reference index, and the running speed can illustrate the performance and practical application value of the algorithm from the side. For the speed evaluation of image detection algorithms, frames per second (FPS) is a commonly used evaluation index, and FPS indicates the number of image frames that can be processed by the algorithm in a unit event, i.e., the number of images that can be processed by the algorithm in one second, which can reflect the running speed of the algorithm more intuitively. Therefore, we use FPS to measure the running speed of the algorithm.

C. Implementation Details

All experiments on the TFDS dataset were run for 300 epochs, with the mechanism of early stopping if the loss did not drop for 40 rounds set. The input image size of the model is 256×256 and the network learning rate is 0.0001. λ_S and λ_G are set to 1. The methodology used in the experiments employs the parameter $n = 3$. The K-value, which is a model parameter discussed as a focus parameter, is preset to (2, 4, 8, 16), which is a multiple K-value combination.

As mentioned in Section II, the K-value represents the size of the pixel block, setting a reasonable K-value can help to improve the model's ability to reconstruct the image, and different K-values in the mask stage also affect the model's learning of the feature information, so we need to validate the effect of K-value on the performance of AD through experiments.

The ablation experiments designed for edge-weighted loss and RBI-mask, the K-value-related experiments, and the quantitative comparison with mainstream models are conducted. The experimental results and discussion are in Section IV.

IV. Discussion

A. Ablation Experiment of Edge-Guided Loss

We use the following metrics to measure the quality of the results.

ROCAUC, the receiver operating characteristic (ROC) curve, is a tool for evaluating the performance of a binary classification model, which characterizes the model's performance by plotting the true class rate (TPR, also known as recall) versus the false positive class rate (FPR) at different threshold settings. AUC refers to the size of the area under the ROC curve, which provides a quantitative model performance metric ranging from 0 to 1, with larger values indicating better classification performance of the model.

FPS is an important metric that measures the performance of a model's real-time image processing system, describing the number of images that the model can process per second.

ACC is the proportion of correctly classified models and is commonly used to assess the ACC of model detection.

We use image- and pixel-level ROCAUC to measure the effectiveness of the detection algorithms. ACC and FPS were also used to evaluate the performance of the model from more perspectives. Table II shows the performance comparison of the model before and after using the edge-guided loss and masks.

TABLE II
Ablation Experiment

Edge- Guide Loss	no	yes	no	yes
Mask	no	no	yes	yes
Image_ROCAUC	0.744	0.852	0.883	0.958
Pixel_ROCAUC	0.746	0.817	0.886	0.924
ACC	0.675	0.822	0.897	0.929
FPS	5.607	5.622	1.813	1.827

TABLE III
Edge Information

TH1	TH2	Edge_ Precision	Image_ ROCAUC	Pixel_ ROCAUC	FPS
25	200	0.494	0.843	0.839	2.955
50	200	0.503	0.901	0.868	2.975
75	200	0.518	0.913	0.857	2.953
100	200	0.521	0.915	0.858	2.964
125	200	0.528	0.947	**0.877**	2.967
150	200	0.516	0.912	0.857	2.957
175	200	0.508	0.871	0.863	2.946
50	400	0.524	0.929	0.859	2.948
100	400	**0.535**	**0.974**	0.865	2.941
150	400	0.516	0.909	0.865	2.960
200	400	0.519	0.829	0.874	**2.996**

It can be seen that the application of both the edge-guided loss and mask mechanisms significantly improves the detection ability of the model, and better detection results can be obtained.

B. Impact of Edge Information Extraction Quality

It is especially critical to study the impact of the quality of edge information on the model detection performance. The edge information extraction module uses the Canny edge detection algorithm. This algorithm regulates the quality of edge extraction through two thresholds. The Edge_Precision of edge information extraction module can be controlled by adjusting different threshold settings. Therefore, to better understand the impact of edge information on model performance, we conducted a control variable experiment on the thresholds of the edge information extraction module. This provides a clearer understanding of how edge information affects model performance. The detection performance of the models trained with different edge information is shown in Table III.

From Table III, it can be seen that Edge_Precision performs best when TH1 = 100 and TH2 = 400, while Image_ROCAUC also performs best. However, it can also be seen that the Edge_Precision does not have much effect on the FPS.

C. Effect of K-Value on Performance

In order to verify the correlation between the pixel size of the component and the K-value, a controlled variable

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TABLE IV
SINGLE K-VALUES

K-value	Image_ROCAUC	Pixel_ROCAUC	FPS
2	0.959	0.939	4.023
4	0.962	0.938	4.470
8	0.972	0.928	4.541
16	0.967	0.926	4.551
32	0.946	0.926	4.506
64	0.885	0.871	4.513
128	0.516	0.777	4.508

TABLE V
DOUBLE K-VALUES

K-value	Image_ROCAUC	Pixel_ROCAUC	FPS
2,4	0.955	0.941	2.849
4,8	0.96	0.939	3.077
8,16	0.974	0.942	3.148
16,32	0.963	0.926	3.146
32,64	0.941	0.914	3.149
64,128	0.533	0.727	3.107

TABLE VI
MULTIPLE K-VALUES

K-value	Image_ROCAUC	Pixel_ROCAUC	FPS
2,4,8	0.945	0.941	2.217
4,8,16	0.938	0.921	2.370
8,16,32	0.948	0.915	2.408
16,32,64	0.961	0.925	2.402
32,64,128	0.563	0.784	2.392
2,4,8,16	0.958	0.924	1.813
4,8,16,32	0.978	0.92	1.932
8,16,32,64	0.966	0.923	1.941
16,32,64,128	0.933	0.929	1.952
2,4,8,16,32	0.931	0.93	1.545
4,8,16,32,64	0.934	0.924	1.614
8,16,32,64,128	0.958	0.928	1.616

TABLE VII
IMAGE_ROCAUC ON DIFFERENT λ_G AND λ_S

λ_G \ λ_S	0.2	0.4	0.6	0.8	1
0.2	0.629	0.793	0.762	0.827	0.889
0.4	0.821	0.791	0.801	0.858	0.781
0.6	0.740	0.761	0.769	0.829	0.732
0.8	0.630	0.801	0.794	0.925	0.800
1	0.779	0.762	0.828	0.845	0.892

experiment was used. Tests were conducted by setting different K-values to reflect the effect of different K-values on the detection performance.

The AD experiments were conducted using a single K-value and the results are shown in Table IV.

By using a single K-value experiment, it can be found that the model ROCAUC and FPS are relatively better when the K-value is 8,16, and 32, and this experimental result verifies our previous speculation.

TABLE VIII
PIXEL_ROCAUC ON DIFFERENT λ_G AND λ_S

λ_G \ λ_S	0.2	0.4	0.6	0.8	1
0.2	0.764	0.829	0.813	0.825	0.784
0.4	0.833	0.817	0.818	0.82	0.893
0.6	0.842	0.817	0.862	0.793	0.817
0.8	0.858	0.779	0.881	0.846	0.919
1	0.907	0.883	0.887	0.891	0.924

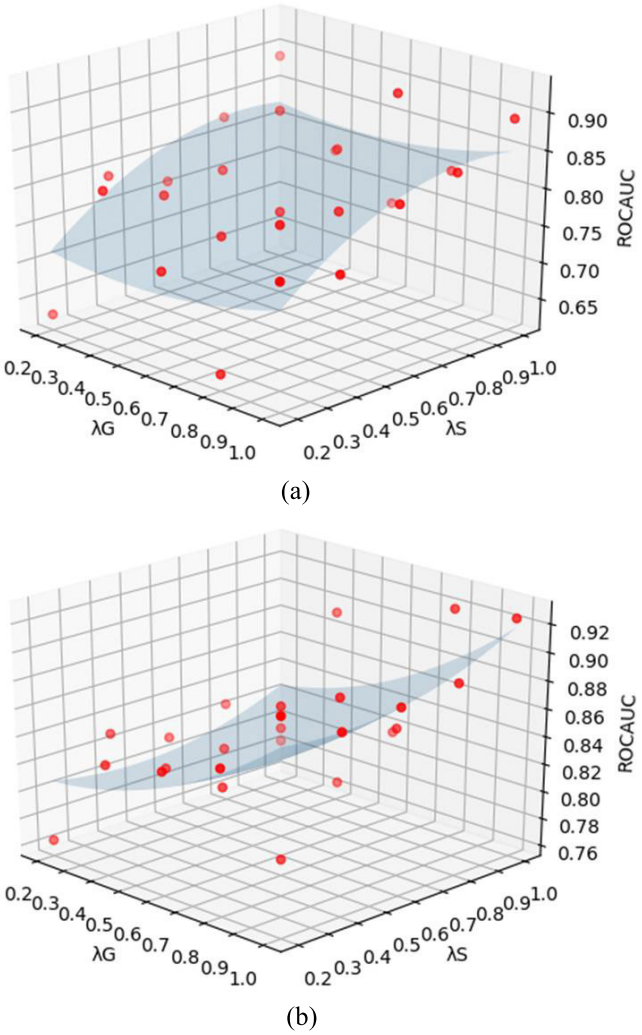


Fig. 7. Fit surface. (a) Image_ROCAUC. (b) Pixel_ROCAUC.

However, the above discussion is based on the common component faults of such anomalies; in practice, in addition to this type of component faults that can be summarized, there are also foreign faults that occur at random locations and abnormal sizes. A single k-value is not a good response to foreign object faults, so the use of multiple k-value combinations can maximize the compatibility of more anomalies and improve the ACC of model detection. Experiments are carried out on double K-values as well as multiple K-values, and the experimental results are shown in Table V.

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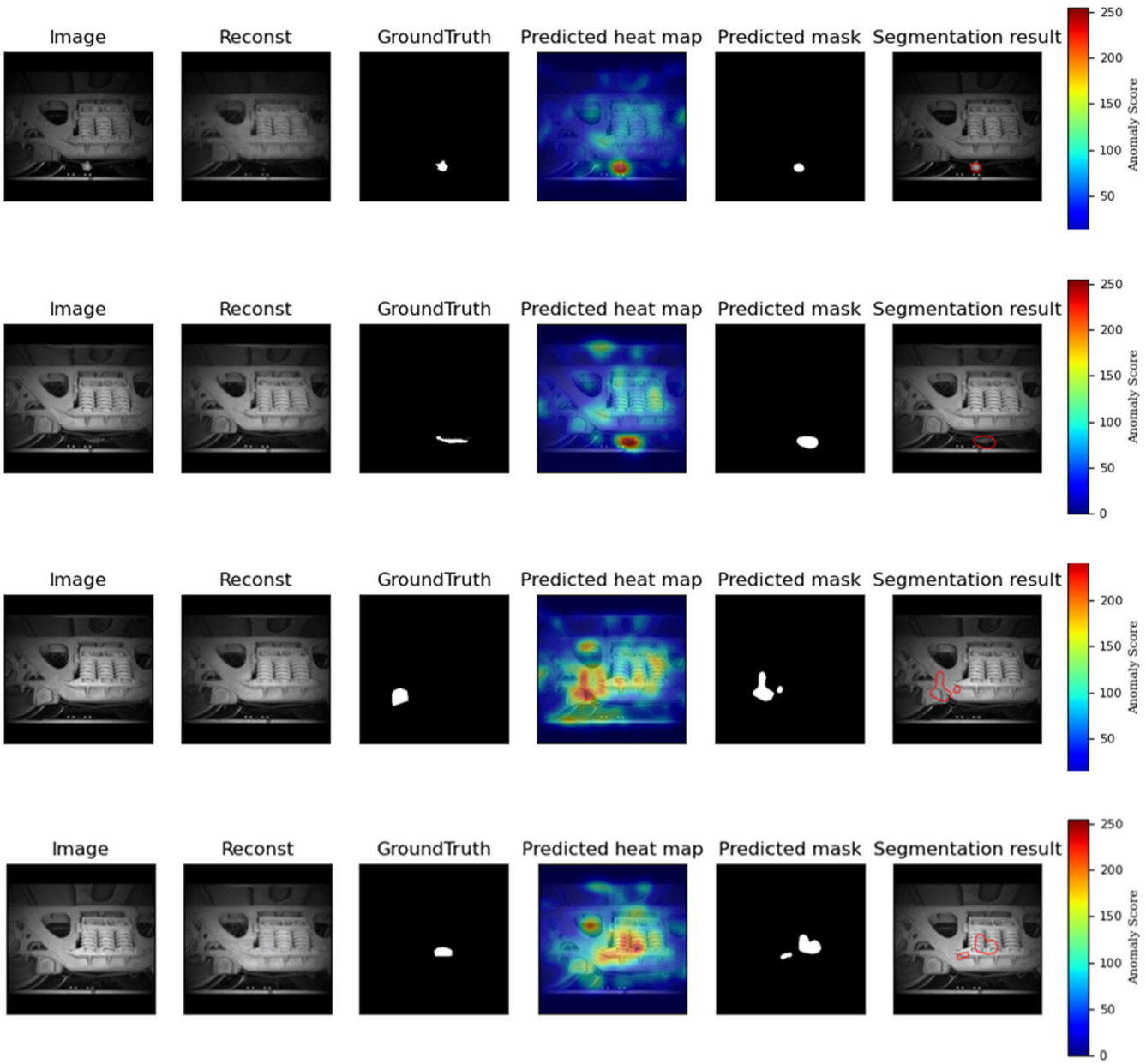


Fig. 8. Case illustrations.

In addition to measuring the performance of AD, the application background of this article also has certain requirements for time. In Tables V and VI, it can be seen that the application of multiple K-values leads to an increase in detection time, and although the model achieves the best detection results for the combination of K-values of (4, 8, 16, 32), the detection takes up more time. When the k-value is taken as (8, 16), the Image_ROCAUC performance is quite good although it is not the best. At the same time, the detection speed can be taken into account and the detection time can be saved. Therefore, we choose K-values of (8, 16) to apply to the model.

D. Effect of λ_G and λ_S on Performance

In the previous section, both λ_G and λ_S are set to 1. Directly setting the weighting coefficient to 1 is for simplicity and also to give equal importance to SSIM and GMS.

In order to verify the effect of the set values of λ_G and λ_S, on the detection performance of the model, experiments with control variables were designed to verify them. The results are shown in Table VII.

The validation shows that the model performance is in the best detection performance range when both parameters are set to 1.

For visual presentation, the data from Table VIII are fit to a surface equation and the surface is shown as follows.

By fitting the surface equations, it is derived that (λ_G, λ_S) corresponds to (1, 0.86569902) for the maximum value of Image_ROCAUC and (1, 1) for the maximum value of Pixel_ROCAUC in the interval [0,1].

It can be verified through experiments that the model detection performance is in the optimal range when λ_G and λ_S are set to 1.

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TABLE IX
ROCAUC AND FPS QUANTITATIVE COMPARISON

Model	ROCAUC	FPS
GANomaly	0.959	2.834
f-AnoGAN	0.966	3.140
VAE(with mask)	0.925	2.794
EdgeAD(ours)	0.974	3.148

E. Quantitative Comparison

Table IX shows the test results of GANomaly, f-AnoGAN, VAE, and the model proposed in this article on the TFDS anomaly dataset.

It can be seen that the proposed method in this article exceeds the existing mainstream AD algorithms in both ROCAUC and FPS, with better detection effects and faster processing speed. From the experimental results, it can be proved that EdgeAD improves efficiency while ensuring a high level of detection ACC. The model is improved in a targeted way according to the TFDS image characteristics and the phenomenon of edge changes caused by the occurrence of anomalies. The new edge information guidance strategy adopted enables the model to capture the edge changes caused by anomalies more accurately and with a certain degree of interpretability through the effective use of edge information.

F. Qualitative Comparison

Fig. 8 shows some examples of the detection results. The results of the AD experiments show that our proposed AD algorithm has a satisfactory effect on TFDS images. By designing a new edge-weighted loss, the image reconstruction model can effectively deal with the characteristics of TFDS images and has better detection results for a variety of common anomalies. It can realize fast and accurate localization and fulfill the needs of practical application scenarios.

V. CONCLUSION

In this article, an AD model based on prior knowledge enhancement is proposed for the problem of intelligent detection of TFDS images, and experimental validation is carried out. By designing the edge-weighted loss function and optimizing the AD process, the model enhances the utilization of image information and effectively improves the detection efficiency of the model. The final experimental results show that the proposed model achieves better performance compared to the current mainstream methods in TFDS AD.

In addition, during the design phase of the model, we targeted the corresponding design to ensure the generalization ability of the model to different anomaly types and scenarios. This provides strong support for the EdgeAD model to have a broad application prospect in practical applications. In our future work, we will further explore and optimize the performance and generalization ability of the EdgeAD model and apply our method to a wide range of AD scenarios.

REFERENCES

[1] Z. Liu, D. Xiao, and Y. Chen, "Displacement fault detection of bearing weight saddle in TFDS based on Hough transform and symmetry validation," in *Proc. 9th Int. Conf. Fuzzy Syst. Knowl. Discovery*, May 2012, pp. 1404–1408, doi: 10.1109/FSKD.2012.6234124.

[2] Y. Cao, B. Zhang, J. Liu, and J. Ma, "Weighted margin sparse embedded classifier for brake cylinder detection," *Neurocomputing*, vol. 120, pp. 560–568, Nov. 2013, doi: 10.1016/j.neucom.2013.04.014.

[3] G. Sun, Y. Zhang, H. Tang, H. Zhang, M. Liu, and D. Zhao, "Railway equipment detection using exact height function shape descriptor based on fast adaptive Markov random field," *Opt. Eng.*, vol. 57, no. 5, p. 1, May 2018, doi: 10.1117/1.oe.57.5.053114.

[4] S. Lu, Z. Liu, and Y. Shen, "Automatic fault detection of multiple targets in railway maintenance based on time-scale normalization," *IEEE Trans. Instrum. Meas.*, vol. 67, no. 4, pp. 849–865, Apr. 2018, doi: 10.1109/TIM.2018.2790498.

[5] M. Karakose, O. Yamanand, K. Murat, and E. Akin, "A new approach for condition monitoring and detection of rail components and rail track in railway," *Int. J. Comput. Intell. Syst.*, vol. 11, no. 1, p. 830, Jan. 2018, doi: 10.2991/ijcis.11.1.63.

[6] J. Sun, Z. Xiao, and Y. Xie, "Automatic multi-fault recognition in TFDS based on convolutional neural network," *Neurocomputing*, vol. 222, pp. 127–136, Jan. 2017, doi: 10.1016/j.neucom.2016.10.018.

[7] X. Fu, K. Li, J. Liu, K. Li, Z. Zeng, and C. Chen, "A two-stage attention aware method for train bearing shed oil inspection based on convolutional neural networks," *Neurocomputing*, vol. 380, pp. 212–224, Mar. 2020, doi: 10.1016/j.neucom.2019.11.002.

[8] L. Xiao, B. Wu, and Y. Hu, "Surface defect detection using image pyramid," *IEEE Sensors J.*, vol. 20, no. 13, pp. 7181–7188, Jul. 2020, doi: 10.1109/JSEN.2020.2977366.

[9] L. Chang, Z. Liu, Y. Shen, and G. Zhang, "Novel multistate fault diagnosis and location method for key components of high-speed trains," *IEEE Trans. Ind. Electron.*, vol. 68, no. 4, pp. 3537–3547, Apr. 2021, doi: 10.1109/TIE.2020.2977580.

[10] L. Zhang, J. Chen, W. Zeng, Z. Tong, Z. Wen, and X. Zhang, "FDNet: Lightweight train image fault detection network in edge computing environments," *IEEE Sensors J.*, vol. 23, no. 20, pp. 25105–25115, Oct. 2023, doi: 10.1109/JSEN.2023.3313155.

[11] T. Ye, Z. Zhang, X. Zhang, Y. Chen, and F. Zhou, "Fault detection of railway freight cars mechanical components based on multi-feature fusion convolutional neural network," *Int. J. Mach. Learn. Cybern.*, vol. 12, no. 6, pp. 1789–1801, Jun. 2021, doi: 10.1007/s13042-021-01274-z.

[12] Y. Zhang, H. Pan, Y. Zhou, M. Li, and G. Sun, "Efficient visual fault detection for freight train braking system via heterogeneous self distillation in the wild," *Adv. Eng. Informat.*, vol. 57, Aug. 2023, Art. no. 102091, doi: 10.1016/j.aei.2023.102091.

[13] N. Hütten, R. Meyes, and T. Meisen, "Vision transformer in industrial visual inspection," *Appl. Sci.*, vol. 12, no. 23, p. 11981, Nov. 2022, doi: 10.3390/app122311981.

[14] Y. Zhang, Y. Zhou, H. Pan, B. Wu, and G. Sun, "Visual fault detection of multiscale key components in freight trains," *IEEE Trans. Ind. Informat.*, vol. 19, no. 8, pp. 9082–9090, Aug. 2023, doi: 10.1109/TII.2022.3224989.

[15] N. Madan et al., "Self-supervised masked convolutional transformer block for anomaly detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 1, pp. 525–542, Jan. 2024, doi: 10.1109/TPAMI.2023.3322604.

[16] X. Liu et al., "Self-supervised learning: Generative or contrastive," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 1, pp. 857–876, Jan. 2023, doi: 10.1109/TKDE.2021.3090866.

[17] I. J. Goodfellow et al., "Generative adversarial networks," *Commun. ACM*, vol. 63, no. 11, pp. 139–144, 2020, doi: 10.1145/3422622.

[18] D. P. Kingma and M. Welling, "Auto-encoding variational Bayes," 2013, *arXiv:1312.6114*.

[19] C. M. V. Neto, L. G. Honorio, and E. P. de Aguiar, "Semi-supervised deep rule-based approach for the classification of wagon bogie springs condition," *Evolving Syst.*, vol. 13, no. 5, pp. 653–666, Oct. 2022, doi: 10.1007/s12530-022-09440-6.

[20] G. Kang, S. Gao, L. Yu, and D. Zhang, "Deep architecture for high-speed railway insulator surface defect detection: Denoising autoencoder with multitask learning," *IEEE Trans. Instrum. Meas.*, vol. 68, no. 8, pp. 2679–2690, Aug. 2019, doi: 10.1109/TIM.2018.2868490.

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[21] H.-D. Zhang et al., "An effective framework using identification and image reconstruction algorithm for train component defect detection," *Appl. Intell.*, vol. 52, no. 9, pp. 10116–10134, Jul. 2022, doi: 10.1007/s10489-021-02981-4.

[22] C. He, H. Shi, J. Si, and J. Li, "Physics-informed interpretable wavelet weight initialization and balanced dynamic adaptive threshold for intelligent fault diagnosis of rolling bearings," *J. Manuf. Syst.*, vol. 70, pp. 579–592, 2023, doi: 10.1016/j.jmsy.2023.08.014.

[23] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Medical Image Computing and Computer-Assisted Intervention—MICCAI 2015*, N. Navab, J. Hornegger, W. M. Wells, and A. F. Frangi, Eds., Cham, Switzerland: Springer, 2015, pp. 234–241.

[24] X. Zhang and K. Zhou, "Multi-period attention for automatic ECG classification," in *Proc. 2nd Int. Conf. Mach. Learn. Comput. Appl. (ICMLCA)*, Shenyang, China, Dec. 2021, pp. 1–4.

[25] V. Zavrtanik, M. Kristan, and D. Skočaj, "Reconstruction by inpainting for visual anomaly detection," *Pattern Recognit.*, vol. 112, Apr. 2021, Art. no. 107706, doi: 10.1016/j.patcog.2020.107706.

[26] Z. Zou, K. Chen, Z. Shi, Y. Guo, and J. Ye, "Object detection in 20 years: A survey," *Proc. IEEE*, vol. 111, no. 3, pp. 257–276, Mar. 2023, doi: 10.1109/JPROC.2023.3238524.

[27] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448, doi: 10.1109/ICCV.2015.169.

[28] W. Xue, L. Zhang, X. Mou, and A. C. Bovik, "Gradient magnitude similarity deviation: A highly efficient perceptual image quality index," *IEEE Trans. Image Process.*, vol. 23, no. 2, pp. 684–695, Feb. 2014, doi: 10.1109/TIP.2013.2293423.

[29] W. Zhou, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004, doi: 10.1109/TIP.2003.819861.

[30] S. Akcay, A. Atapour-Abarghouei, and T. P. Breckon, "GANomaly: Semi-supervised anomaly detection via adversarial training," in *Computer Vision—ACCV 2018*, C. V. Jawahar, H. Li, G. Mori, and K. Schindler, Eds., Cham, Switzerland: Springer, 2019, pp. 622–637.

[31] T. Schlegl, P. Seeböck, S. M. Waldstein, G. Langs, and U. Schmidt-Erfurth, "F-AnoGAN: Fast unsupervised anomaly detection with generative adversarial networks," *Med. Image Anal.*, vol. 54, pp. 30–44, May 2019, doi: 10.1016/j.media.2019.01.010.

[32] S. Akçay, A. Atapour-Abarghouei, and T. P. Breckon, "Skip-GANomaly: Skip connected and adversarially trained encoder–decoder anomaly detection," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Budapest, Hungary, Jul. 2019, pp. 1–8.

[33] T. N. Nguyen and J. Meunier, "Anomaly detection in video sequence with appearance-motion correspondence," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Seoul, South Korea, Oct. 2019, pp. 1273–1283, doi: 10.1109/ICCV.2019.00136.

[34] R. T. Ionescu, F. S. Khan, M.-I. Georgescu, and L. Shao, "Object-centric auto-encoders and dummy anomalies for abnormal event detection in video," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 7834–7843, doi: 10.1109/CVPR.2019.00803.



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